

GMUNDEN, Austria

WMO: 111540

Lat: 47.915N

Lon: 13.807E

Elev: 1404

StdP: 13.97

Time Zone: 1.00 (EUC)

Period: 94-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.9	16.8	9.3	9.3	13.8	13.6	11.5	18.7	16.5	48.3	14.5	45.2	2.6	20	0.455

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.5	87.0	70.0	83.5	68.4	80.3	66.8	71.1	84.2	69.3	81.4	67.8	78.2	3.8	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.7	103.5	77.1	65.3	98.4	74.8	63.9	93.6	73.0	35.8	84.7	34.3	81.6	32.9	78.1	78.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.2	10.4	9.0	10.0	92.9	6.0	3.2	5.7	95.3	2.2	97.2	-1.2	99.0	-5.5	101.3		
(5)			9.5	74.2	5.8	2.4	5.3	75.9	1.9	77.4	-1.3	78.7	-5.5	80.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.1	31.7	35.1	41.8	50.1	58.1	63.8	67.8	67.3	58.8	50.1	41.0	34.3		
	DBStd	14.50	8.48	8.29	7.36	7.47	6.72	7.04	6.08	6.13	5.96	6.96	7.16	7.99		
	HDD50	2215	568	419	267	90	10	1	0	0	4	87	281	488		
	HDD65	5793	1032	838	719	448	231	106	40	45	200	462	720	952		
	CDD50	2245	1	1	13	93	262	416	553	536	268	90	11	1		
	CDD65	349	0	0	0	1	18	71	128	116	14	1	0	0		
	CDH74	2611	0	0	0	19	178	570	942	824	77	1	0	0		
	CDH80	759	0	0	0	2	27	154	312	254	10	0	0	0		
	WSAvg	3.6	4.4	4.2	4.1	3.5	3.4	3.4	3.6	3.2	3.0	2.8	3.5	3.8		
	PrecAvg	57.0	3.8	3.1	4.4	3.5	5.3	6.5	6.8	6.3	5.4	3.9	4.0	4.0		
	PrecMax	68.5	8.7	7.1	8.1	7.1	10.7	13.1	13.9	13.5	10.4	6.9	9.7	8.0		
(15) Precipitation	PrecMin	44.9	0.3	1.0	1.8	0.8	1.2	3.3	2.4	2.4	2.0	1.2	0.4	1.3		
	PrecStd	5.6	2.0	1.6	2.1	1.5	2.1	2.0	2.6	2.2	2.2	1.6	1.9	1.6		
	DB	55.4	57.3	66.0	76.4	83.5	88.8	92.0	91.8	80.8	71.3	64.4	54.2			
	MCWB	47.1	46.8	52.4	59.7	66.8	71.1	72.9	70.4	66.9	60.5	52.5	45.9			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	50.3	52.4	60.9	71.6	79.2	84.6	87.1	86.4	75.7	66.2	57.8	51.0			
	2% MCWB	44.8	45.2	49.9	57.4	64.6	69.3	70.5	69.1	64.2	57.9	50.4	45.4			
	5% DB	46.7	49.0	56.5	67.4	75.0	80.9	83.3	82.3	71.9	62.7	53.8	47.7			
	5% MCWB	42.5	43.2	47.9	55.1	61.8	66.9	68.6	67.9	62.5	56.3	49.6	43.5			
	10% DB	43.4	45.6	52.4	63.3	70.9	76.8	79.3	78.6	68.2	59.7	50.7	44.8			
	10% MCWB	39.9	41.2	45.8	52.7	59.4	64.5	66.9	66.3	60.6	54.9	47.2	41.5			
	0.4% WB	48.1	48.6	53.5	61.1	68.3	72.5	74.5	72.5	68.3	61.9	53.6	48.3			
	0.4% MCDB	53.4	55.1	63.8	74.9	81.4	86.0	89.3	86.3	77.9	68.6	60.8	52.5			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	45.3	45.9	50.7	57.9	65.5	70.2	71.6	70.3	65.5	59.2	51.4	46.0			
	2% MCDB	49.8	51.8	59.0	70.4	77.3	82.6	84.6	83.4	73.4	64.1	56.1	49.8			
	5% WB	42.5	43.7	48.6	55.6	62.9	68.1	69.5	68.7	63.5	57.2	49.6	43.8			
	5% MCDB	46.3	48.3	55.3	66.3	72.8	78.7	81.6	80.3	70.1	61.4	53.8	47.2			
	10% WB	40.0	41.6	46.5	53.2	60.5	65.7	67.6	67.2	61.6	55.4	47.5	41.5			
	10% MCDB	43.2	45.3	51.9	61.9	69.5	74.4	77.3	77.0	67.0	59.3	50.8	44.9			
	MDBR	8.2	10.6	13.8	16.6	16.8	16.1	16.5	16.4	13.7	12.5	8.8	8.2			
	5% DB	11.9	15.9	21.7	25.6	25.3	23.9	24.1	23.4	20.1	17.1	14.7	12.2			
(36) Mean Daily Temperature Range	MCDBR	8.0	9.9	12.3	13.6	12.3	10.5	10.0	9.0	9.7	9.6	9.2	8.0			
	MCWBR	11.1	14.7	20.4	24.5	24.0	22.5	22.5	21.7	18.4	15.0	13.4	11.8			
	MCWBR	8.0	9.6	12.1	13.5	12.0	10.4	10.1	9.1	9.5	9.0	8.6	8.4			
	MCWBR	8.0	9.6	12.1	13.5	12.0	10.4	10.1	9.1	9.5	9.0	8.6	8.4			
(40) Clear-Sky Solar Irradiance	taub	0.295	0.323	0.364	0.399	0.409	0.414	0.419	0.411	0.383	0.350	0.322	0.294			
	taud	2.401	2.330	2.286	2.253	2.265	2.292	2.278	2.318	2.371	2.442	2.466	2.442			
	Ebn at Noon	248	264	269	270	271	270	266	263	260	251	236	234			
	Edh at Noon	21	29	35	40	41	40	40	37	32	26	20	18			
(44) All-Sky Solar Radiation	RadAvg	365	656	979	1329	1531	1684	1669	1482	1070	716	387	291			
	RadStd	49	84	116	162	181	157	161	167	141	75	54	41			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (99 neighbors)	+0.90	N/A	N/A	+1.35	+0.76	+0.65	N/A	N/A	+135	+61

Nomenclature: See separate page